

$Z\varphi$ -classes for three perfect groups of order 1920

PerfectGroup(1920, 5), PerfectGroup(1920, 6) and PerfectGroup(1920, 7)

Quotient orders $2 \leq n \leq 127$

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Important interpretation. The table is *not* a cumulative count over the three perfect groups. For a fixed n , the same number of extensions is obtained independently for each of the three pairwise non-isomorphic perfect residuals P_5, P_6, P_7 . Consequently, if $T(n)$ denotes the cumulative number in the table for one fixed P_i , then the total number of nonsolvable groups arising from these three perfect residuals, with quotient order at most n , is $3T(n)$. In particular, since $T(127) = 123\,338$, the number of such nonsolvable groups of order at most $1920 \cdot 127 = 243\,840$ is

$$\boxed{3 \cdot 123\,338 = 370\,014}.$$

This count concerns the extensions with quotient orders $2 \leq n \leq 127$.

Main table of $Z\varphi$ -classes

n	$ S $	#SG	central	nontrivial	classes/ P_i	cumulative/ P_i	$3\times$ cumulative
2	4	2	2	2	4	4	12
3	6	2	1	0	1	5	15
4	8	5	6	8	14	19	57
5	10	2	1	0	1	20	60
6	12	5	4	4	8	28	84
7	14	2	1	0	1	29	87
8	16	14	21	42	63	92	276
9	18	5	2	0	2	94	282
10	20	5	4	4	8	102	306
11	22	2	1	0	1	103	309
12	24	15	16	24	40	143	429
13	26	2	1	0	1	144	432
14	28	4	4	4	8	152	456
15	30	4	1	0	1	153	459
16	32	51	95	286	381	534	1602
17	34	2	1	0	1	535	1605
18	36	14	10	10	20	555	1665
19	38	2	1	0	1	556	1668

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Main table of $Z\varphi$ -classes – continued

n	$ S $	#SG	central	nontrivial	classes/ P_i	cumulative/ P_i	$3\times$ cumulative
20	40	14	16	26	42	598	1794
21	42	6	2	0	2	600	1800
22	44	4	4	4	8	608	1824
23	46	2	1	0	1	609	1827
24	48	52	72	166	238	847	2541
25	50	5	2	0	2	849	2547
26	52	5	4	4	8	857	2571
27	54	15	5	0	5	862	2586
28	56	13	14	24	38	900	2700
29	58	2	1	0	1	901	2703
30	60	13	8	8	16	917	2751
31	62	2	1	0	1	918	2754
32	64	267	649	3026	3675	4593	13779
33	66	4	1	0	1	4594	13782
34	68	5	4	4	8	4602	13806
35	70	4	1	0	1	4603	13809
36	72	50	48	80	128	4731	14193
37	74	2	1	0	1	4732	14196
38	76	4	4	4	8	4740	14220

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Main table of $Z\varphi$ -classes – continued

n	$ S $	#SG	central	nontrivial	classes/ P_i	cumulative/ P_i	$3\times$ cumulative
39	78	6	2	0	2	4742	14226
40	80	52	73	176	249	4991	14973
41	82	2	1	0	1	4992	14976
42	84	15	12	12	24	5016	15048
43	86	2	1	0	1	5017	15051
44	88	12	14	24	38	5055	15165
45	90	10	2	0	2	5057	15171
46	92	4	4	4	8	5065	15195
47	94	2	1	0	1	5066	15198
48	96	231	422	1574	1996	7062	21186
49	98	5	2	0	2	7064	21192
50	100	16	10	10	20	7084	21252
51	102	4	1	0	1	7085	21255
52	104	14	16	26	42	7127	21381
53	106	2	1	0	1	7128	21384
54	108	45	30	30	60	7188	21564
55	110	6	2	0	2	7190	21570
56	112	43	64	158	222	7412	22236
57	114	6	2	0	2	7414	22242

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Main table of $Z\varphi$ -classes – continued

n	$ S $	#SG	central	nontrivial	classes/ P_i	cumulative/ P_i	$3\times$ cumulative
58	116	5	4	4	8	7422	22266
59	118	2	1	0	1	7423	22269
60	120	47	46	84	130	7553	22659
61	122	2	1	0	1	7554	22662
62	124	4	4	4	8	7562	22686
63	126	16	4	0	4	7566	22698
64	128	2328	8308	72010	80318	87884	263652
65	130	4	1	0	1	87885	263655
66	132	10	8	8	16	87901	263703
67	134	2	1	0	1	87902	263706
68	136	15	16	26	42	87944	263832
69	138	4	1	0	1	87945	263835
70	140	11	8	8	16	87961	263883
71	142	2	1	0	1	87962	263886
72	144	197	262	655	917	88879	266637
73	146	2	1	0	1	88880	266640
74	148	5	4	4	8	88888	266664
75	150	13	3	0	3	88891	266673
76	152	12	14	24	38	88929	266787

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Main table of $Z\varphi$ -classes – continued

n	$ S $	#SG	central	nontrivial	classes/ P_i	cumulative/ P_i	$3\times$ cumulative
77	154	4	1	0	1	88930	266790
78	156	18	12	12	24	88954	266862
79	158	2	1	0	1	88955	266865
80	160	238	445	1664	2109	91064	273192
81	162	55	15	0	15	91079	273237
82	164	5	4	4	8	91087	273261
83	166	2	1	0	1	91088	273264
84	168	57	56	104	160	91248	273744
85	170	4	1	0	1	91249	273747
86	172	4	4	4	8	91257	273771
87	174	4	1	0	1	91258	273774
88	176	42	63	158	221	91479	274437
89	178	2	1	0	1	91480	274440
90	180	37	20	20	40	91520	274560
91	182	4	1	0	1	91521	274563
92	184	12	14	24	38	91559	274677
93	186	6	2	0	2	91561	274683
94	188	4	4	4	8	91569	274707
95	190	4	1	0	1	91570	274710

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Main table of $Z\varphi$ -classes – continued

n	$ S $	#SG	central	nontrivial	classes/ P_i	cumulative/ P_i	$3\times$ cumulative
96	192	1543	3914	23798	27712	119282	357846
97	194	2	1	0	1	119283	357849
98	196	12	10	10	20	119303	357909
99	198	10	2	0	2	119305	357915
100	200	52	52	88	140	119445	358335
101	202	2	1	0	1	119446	358338
102	204	12	8	8	16	119462	358386
103	206	2	1	0	1	119463	358389
104	208	51	73	176	249	119712	359136
105	210	12	2	0	2	119714	359142
106	212	5	4	4	8	119722	359166
107	214	2	1	0	1	119723	359169
108	216	177	162	288	450	120173	360519
109	218	2	1	0	1	120174	360522
110	220	15	12	12	24	120198	360594
111	222	6	2	0	2	120200	360600
112	224	197	383	1520	1903	122103	366309
113	226	2	1	0	1	122104	366312
114	228	15	12	12	24	122128	366384

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Main table of $Z\varphi$ -classes – continued

n	$ S $	#SG	central	nontrivial	classes/ P_i	cumulative/ P_i	$3\times$ cumulative
115	230	4	1	0	1	122129	366387
116	232	14	16	26	42	122171	366513
117	234	16	4	0	4	122175	366525
118	236	4	4	4	8	122183	366549
119	238	4	1	0	1	122184	366552
120	240	208	275	760	1035	123219	369657
121	242	5	2	0	2	123221	369663
122	244	5	4	4	8	123229	369687
123	246	4	1	0	1	123230	369690
124	248	12	14	24	38	123268	369804
125	250	15	5	0	5	123273	369819
126	252	46	32	32	64	123337	370011
127	254	2	1	0	1	123338	370014

Runtime concentration

Values of n accounting together for more than 90% of the total runtime

rank	n	$ S $	#SG	runtime (s)	cumulative share
1	64	128	2328	615.358	39.636%
2	96	192	1543	513.186	72.692%
3	120	240	208	58.975	76.491%
4	112	224	197	54.215	79.983%
5	48	96	231	49.493	83.171%
6	80	160	238	47.444	86.227%
7	108	216	177	38.929	88.734%
8	72	144	197	35.747	91.037%

These eight quotient orders account for more than 90% of the measured runtime. In particular, $n = 64$ and $n = 96$ alone account for approximately 72.7% of the total computation time.

Abstract

For each of the three perfect groups

$$P_i = \text{PerfectGroup}(1920, i), \quad i \in \{5, 6, 7\},$$

we have $Z(P_i) \cong C_2$, $\text{Out}(P_i) \cong C_2$, and the automorphism extension splits over $\text{Inn}(P_i)$. The $Z\varphi$ method therefore leads to the same counting problem for each of the three perfect residuals. For a solvable quotient R of order n , the auxiliary group S has order $2n$, and the number in the column “classes/ P_i ” is the number of non-isomorphic extensions for *each one* of P_5 , P_6 , and P_7 .

Computational method

The computations were performed in GAP using the optimized **TwoFusions** implementation of the $Z\varphi$ method. For each n , all groups $S = \text{SmallGroup}(2n, j)$ are processed. The central contribution is the sum of **fusz2**, and the nontrivial coupling contribution is the sum of **fuspair**. Their sum gives the number of $Z\varphi$ -classes for one fixed perfect residual.

The computation is performed only once for every n : its result applies identically and independently to P_5 , P_6 , and P_7 . The factor three is introduced only when counting the union of the three disjoint extension families.

Final totals

For every one of the three perfect groups separately,

$$\sum_{n=2}^{127} N(P_i, n) = 123\,338.$$

Since the three perfect residuals are pairwise non-isomorphic, their extension families are disjoint. Therefore

$$\sum_{i \in \{5,6,7\}} \sum_{n=2}^{127} N(P_i, n) = 3 \cdot 123\,338 = \boxed{370\,014}.$$

The maximum group order represented here is

$$1920 \cdot 127 = 243\,840.$$

The total measured runtime for the common computation was 1552.505 seconds.

Reproducibility

For quotient order n , the program enumerates all **SmallGroups** of order $2n$ and applies the same **TwoFusions** computation. The table records the common count for one perfect residual; it does not add the three perfect groups together. Computations, data and presentation: Claude Archer, 2026.